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#	Modül	Öğreneceğin konular	Çıktı
1	İş Problemi & KPI	KPI vs metric, revenue, profit, margin, growth, conversion, churn, retention, funnel, north-star metric	İş problemini analitik probleme çevirebilmek
2	Python Temelleri	syntax, değişkenler, data types, loops, functions, comprehensions, modules, exceptions, OOP temelleri	Python ile rahat kod yazmak
3	NumPy	arrays, indexing, slicing, broadcasting, vectorization, statistics, linear algebra operations	Sayısal hesaplama
4	Pandas	Series, DataFrame, filtering, groupby, merge, pivot, reshape, datetime, missing values	Gerçek datasetleri işlemek
5	SQL	SELECT, WHERE, JOIN, GROUP	Veritabanından analitik veri çekmek

		BY, HAVING, CASE, CTE, subquery, window functions, date functions	
6	Veri Temizleme	null, duplicate, invalid values, outliers, encoding, types, data quality kontrolleri	Raw → clean dataset
7	Exploratory Data Analysis	distributions, segmentation, correlations, anomaly analysis, cohort analysis	Veriyi modelden önce anlamak
8	Data Visualization	Matplotlib, Plotly, Power BI/Tableau, chart selection, storytelling	Analizleri anlatabilmek
9	Olasılık	conditional probability, Bayes, random variables, expected value, common distributions	İstatistik altyapısı
10	İstatistik	mean/ median/ variance, sampling, CLT, confidence intervals, hypothesis tests	İstatistiksel çıkarım

11 A/B Testing	experiment design, H0/H1, p-value, alpha/beta, power, sample size, Type I/II errors	Deney sonuçlarını değerlendirmek
12 Lineer Cebir	vectors, matrices, dot product, matrix multiplication, eigenvalues/eigenvectors	ML matematiğini anlamak
13 Calculus & Optimization	derivatives, partial derivatives, gradient, gradient descent, loss minimization	Model optimizasyonunu anlamak
14 Machine Learning Temelleri	supervised/unsupervised learning, train/validation/test, overfitting, underfitting, bias/variance	ML düşünce yapısı
15 Regression	Linear Regression, Ridge, Lasso, ElasticNet, polynomial regression	Sayısal tahmin
16 Classification	Logistic Regression, KNN, Naive Bayes, Decision Tree	Sınıflandırma
17 Ensemble Models	Random Forest, Gradient Boosting, XGBoost,	Güçlü tabular modeller

18 Model Evaluation	LightGBM, CatBoost MAE, RMSE, R^2, accuracy, precision, recall, F1, ROC-AUC, PR-AUC, confusion matrix	Doğru metriği seçmek
19 Feature Engineering	transformations, binning, encoding, scaling, interactions, domain features	Model performansını artırmak
20 ML Pipelines	Pipeline, ColumnTransformer, preprocessing, reproducibility	Leakage'siz ML pipeline
21 Cross Validation	KFold, StratifiedKFold, TimeSeriesSplit	Güvenilir model karşılaştırması
22 Hyperparameter Optimization	GridSearchCV, RandomizedSearchCV, Optuna mantığı	Model tuning
23 Imbalanced Data	class weights, over/undersampling, SMOTE, threshold tuning	Dengesiz veri problemleri
24 Explainable AI	feature importance, permutation importance, SHAP, PDP	Modeli açıklayabilmek
25 Clustering	K-Means, hierarchical clustering,	Segmentasyon

		DBSCAN	
26	Dimensionality Reduction	PCA, t-SNE/UMAP kavramları	Yüksek boyutlu veriyi analiz etmek
27	Time Series	trend, seasonality, lag, rolling, ARIMA, Prophet yaklaşımı, ML forecasting	Forecasting modelleri
28	NLP	text cleaning, tokenization, TF-IDF, embeddings, classification	Metin analizi
29	Deep Learning	neural networks, activation, forward/backpropagation, optimizers, regularization	Neural network temeli
30	PyTorch	tensors, Dataset/DataLoader, model, loss, optimizer, training loop	DL modelleri geliştirmek
31	Transformers & LLM	attention, transformer, embeddings, Hugging Face, prompting, RAG	Modern AI uygulamaları
32	Vector Databases	embeddings, similarity search, FAISS/Chroma/Pinecone mantığı	Semantic search/RAG
33	Recommendation Systems	collaborative filtering,	Öneri sistemi

	content-based, ranking	
34 Data Engineering Temelleri	ETL/ELT, pipelines, batch vs streaming, data warehouse/lake	Veri akışını anlamak
35 Spark	distributed computing, DataFrame API, PySpark	Büyük veri işleme
36 Cloud	AWS/ Azure/GCP temel servisleri, storage, compute, database, ML services	Cloud üzerinde proje çalıştırmak
37 Git & GitHub	commit, branch, merge, pull request, issues, repo yapısı	Profesyonel proje yönetimi
38 Software Engineering	clean code, project structure, logging, configuration, tests, type hints	Notebook dışına çıkmak
39 API Development	FastAPI, REST, JSON, validation	ML modelini servis etmek
40 Docker	images, containers, Dockerfile, volumes, environment variables	Uygulamayı paketlemek
41 MLOps	experiment tracking, MLflow, model registry, versioning	Model yaşam döngüsü

42 Deployment	API deployment, batch inference, cloud deployment	Modeli production'a taşımak
43 Monitoring	data drift, concept drift, performance monitoring, retraining	Production modelini izlemek
44 Data Ethics & Governance	bias, fairness, privacy, leakage, explainability, security	Güvenilir modeller geliştirmek
45 Data Storytelling	executive summary, business recommendations, presentation	Teknik sonucu iş kararına çevirmek
46 Portfolio & GitHub	README, reproducibility, architecture, notebooks, source code, demo	İş başvurusu portföyü
47 Data Science Interview	Python, SQL, statistics, ML theory, case study, experiment design	Mülakata hazırlanmak
48 System Design for ML	batch/online inference, feature pipelines, model serving, scalability	Senior seviyeye giden altyapı